HireFlow

A recruitment tracker deployed through a verified DevSecOps pipeline

FastAPI + React + PostgreSQL

GitHub Actions > GHCR > Argo CD > Amazon EKS

Repository: github.com/Jagrit3500/DevOps_Capstone_Project

Project overview

A working candidate pipeline rather than a static demo page

- Create, update, filter, and delete candidates across hiring stages.
- The React dashboard calls the FastAPI backend through /api.
- PostgreSQL stores candidate data on an EBS persistent volume.
- Demo data is opt-in locally; the cloud deployment starts empty.

Technology stack

Each layer has a defined responsibility

- React and Vite build the frontend; Nginx serves it as a non-root user.
- FastAPI, SQLAlchemy, Alembic, and PostgreSQL run the API and storage.
- Docker packages both apps; GitHub Actions tests, scans, and publishes them.
- Terraform creates AWS networking and EKS; Helm describes workloads.
- Argo CD applies Git state; Prometheus and Grafana show live metrics.

Architecture

The deployed request and delivery paths

DELIVERY PATH



EKS pulls the SHA-tagged application images from GHCR.

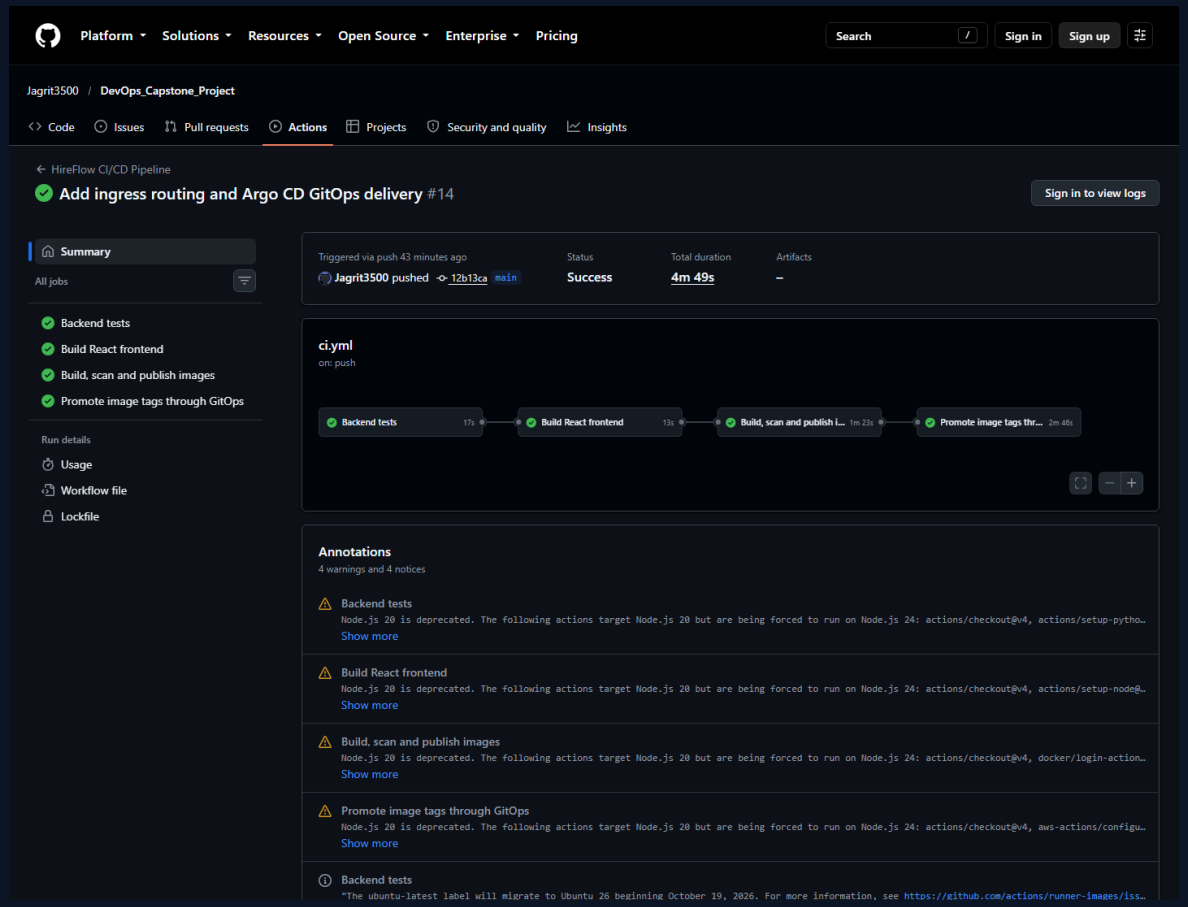
REQUEST PATH



Ingress sends / to frontend and /api to backend. Only the backend accesses PostgreSQL.

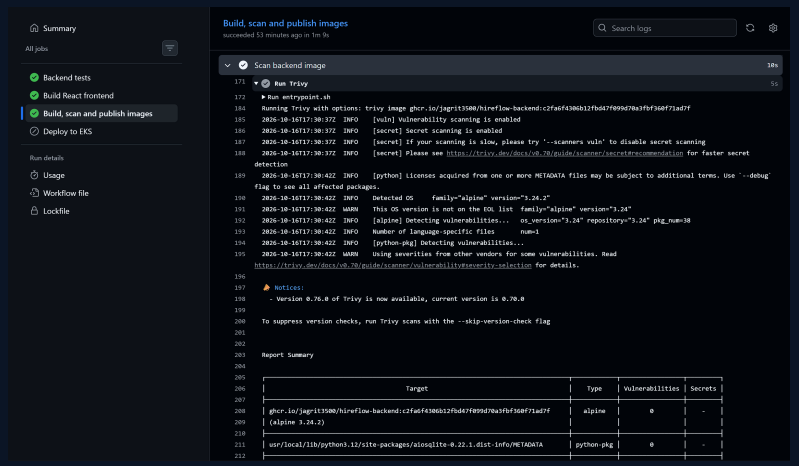
CI/CD pipeline

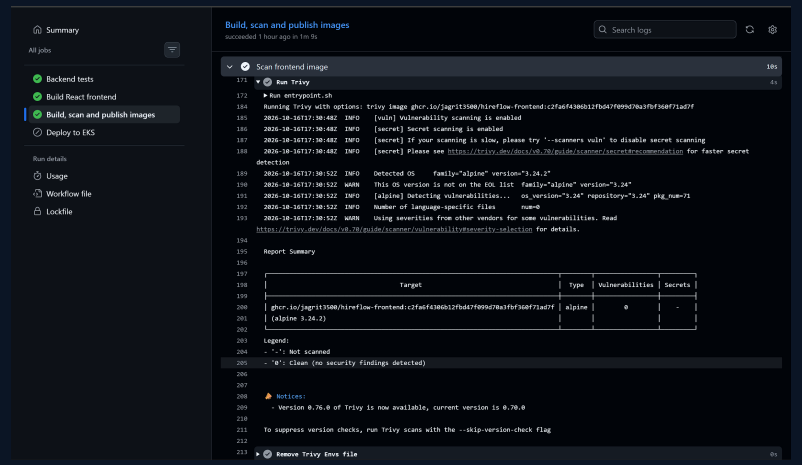
Run 39343899906 completed successfully on the GitOps change



DevSecOps gates

Both container images passed the HIGH and CRITICAL Trivy checks





No application password or AWS access key is committed to Git.

Workloads read the database password from a Kubernetes Secret.

Infrastructure

Terraform provisioned the Mumbai EKS environment

- VPC with two public and two private subnets across two Availability Zones.
- NAT and Internet Gateways, IAM roles, EBS CSI, and Metrics Server.
- EKS 1.35 runs two Ready c7i-flex.large workers in private subnets.
- Terraform fmt and validate passed; the latest full-refresh plan had no drift.

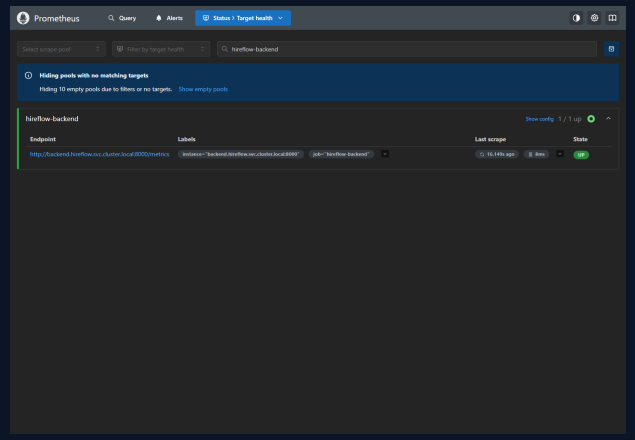
Kubernetes and GitOps

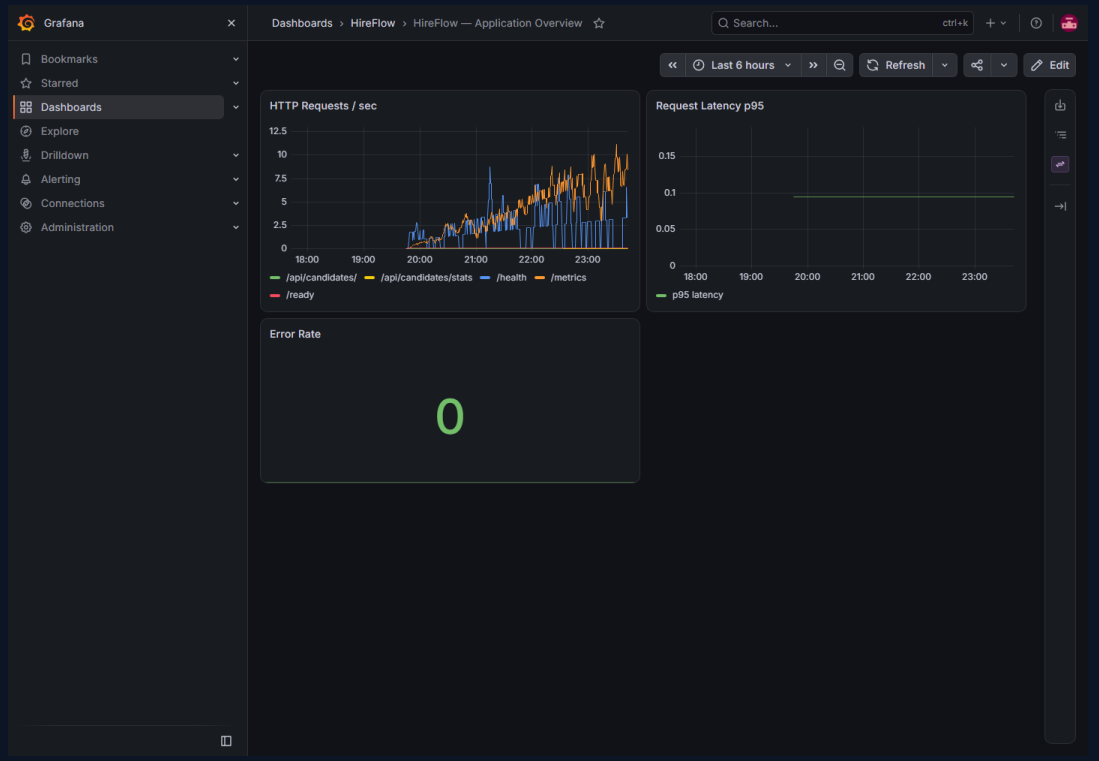
Argo CD reports Synced and Healthy at commit c5093b6

- Backend and frontend Deployments each have 2/2 available replicas.
- PostgreSQL StatefulSet is 1/1 ready with a bound EBS claim.
- Both app Services are ClusterIP; ingress-nginx exposes / and /api.
- HPA has live CPU metrics from the EKS Metrics Server.

Observability

Prometheus targets and Grafana panels contain live data





Demo and lessons

What the evidence proves and what comes after evaluation

- Public Ingress returned HTTP 200 for /, healthy JSON for /health, and [] for /api/candidates/.
- A failing PostgreSQL initdb needed PGDATA inside the mounted volume.
- Short-lived AWS login and intermittent DNS required careful reruns.
- The live stack incurs AWS costs. Teardown belongs after the reviewer sees it.